Signed, Sealed, Delivered: Digital Receipts in the Ugandan Dairy Chain*

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Abstract

We provide causal evidence that digital receipts—SMS messages reporting milk deliveries—can improve accountability, delivery behavior, and product quality in agricultural markets. In a randomized experiment with dairy cooperatives in western Uganda, we find that treated farmers, particularly those who self-deliver, are significantly more likely to deliver milk. This increase in delivery activity occurs without a change in average volumes delivered, suggesting a shift along the extensive margin. The intervention also improves milk quality: treated farmers deliver higher-quality (less diluted) milk, as measured by lactometer readings. Among farmers who rely on transporters, digital receipts increase the likelihood of detecting discrepancies, switching baseline transporters, and reporting lower trust in intermediaries. These results show that the intervention works differently depending on whether information frictions are present: for self-deliverers, who already have full observability over their deliveries, impacts are concentrated on participation, while for transporter users, where observability is limited, impacts are concentrated on detecting discrepancies and moving away from dishonest transporters. By turning unobservable transactions into verifiable digital records, SMS receipts enhance transparency and make intermediary behavior more observable. Overall, our findings show how simple digital tools can reduce information frictions, shift farmer and intermediary behavior, and improve product quality in fragmented agricultural systems.

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1 Introduction

In many agricultural markets, especially in low- and middle-income countries, smallholder farmers face barriers to receiving fair compensation for their products and accessing larger markets. As a result, cooperatives are common in agricultural value chains in these settings, where farmers pool their output and sell collectively to increase their bargaining power. Depending on distance and transportation options, some farmers deliver their products themselves, while others rely on intermediaries to transport and sell their goods, often without visibility into what happens after collection. These information frictions open the door to inefficiencies and opportunistic behavior, and can ultimately lead to market failures, deviating from the idea of perfectly informed and efficient markets (Arrow & Debreu, 1954; Hayek, 1945; Smith, 1776). This problem is particularly pronounced in Uganda's dairy sector, where smallholder farmers depend on informal transporters to deliver milk to cooperatives. These transporters typically hold key informational advantages: they observe what happens to the milk after collection, including any dilution and the actual quantity delivered. Such asymmetries create opportunities for strategic behavior and misreporting, ultimately undermining farmers' earnings and weakening trust in the supply chain.

For instance, transporters may under-report milk volumes, reducing the amount credited to farmers. They can also siphon off milk from containers and deliver it to whichever collection center (including those run by other cooperatives) offers cash payments. Additionally, transporters may dilute milk with water to artificially inflate volume, profiting from the additional liters, by siphoning them off, while harming milk quality. Since farmers are paid biweekly in lump sums, prices change frequently, and they lack timely data on deliveries, such discrepancies are difficult to detect and harder to punish, making formal contract enforcement largely ineffective.

While the delivery decision is made by the farmer, its timing differs fundamentally depending on how the farmer delivers their milk. Farmers who self-deliver make an active decision each day about whether to deliver, and if so, how much to deliver, taking into account their transportation constraints. In contrast, farmers who rely on transporters face two key constraints. First, the decision to deliver must be made in advance, since pick-up frequency and timing need to be scheduled ahead of time. Second, the volume they can send is limited by the transporter's available capacity at the time of pick-up. While both types of farmers ultimately make the decision on whether and how much to deliver, the timing of the decision differs—making delivery behavior among farmers who use transporters somewhat more *sticky* compared to those who self-deliver.

Building on Holmström (1979) and Levin (2003), this paper examines whether improving transparency, by making post-harvest transactions more observable through a digital tool, can mitigate information frictions in a context where relational contracts under low monitoring are prevalent. We study the introduction of a digital receipt system that sends farmers SMS messages reporting the daily quantities of milk delivered and recorded at the cooperative on their behalf. By reducing information asymmetries, the intervention was designed to improve farmers' ability to monitor transporter behavior, adjust delivery arrangements if needed, and potentially increase milk deliveries and reduce opportunities for quality-related misconduct. Our work complements recent evidence from Fabregas et al. (2019), who show that SMS-based interventions can shape farmer behavior by improving access to timely and actionable information. Finally, it speaks to findings by Saitone et al. (2018), who document how payment delays and limited accountability contribute to farmers' hesitancy to sell through cooperatives—barriers that digital receipts may help to address.

We collected high-frequency administrative data from cooperatives, including daily milk prices, volumes per farmer, revenues per farmer, transporter used per farmer, and milk quality at the farmer level. We complement administrative records with surveys from farmers and transporters.

We have five main findings: First, digital receipts improved milk quality, with treated farmers delivering milk with lower water content. Second, treated farmers who use transporters were 20–26 percentage points more likely to detect discrepancies compared to non-treated farmers using transporters. Third, treated farmers who use transporters were 14–17 percentage points more likely to change their baseline transporter. Fourth, treated farmers who use transporters were also less likely to trust their current transporter compared to non-treated farmers using transporters. Fifth, while total delivered volumes did not increase, treated farmers were more likely to deliver during the rainy season, mainly those who self-deliver. Additionally, we observe a higher share of farmers in our sample delegating their deliveries to a transporter. Overall, the findings show that improving observability through digital receipts can shift farmer behavior along several margins—even without price premiums or introducing formal enforcement. The intervention helped farmers better monitor transporter behavior, increased accountability, and led many to adjust their delivery decisions. While we don't find an increase in total volumes, we do see changes in delivery frequency and transporter switching, suggesting that better information can help farmers navigate these markets more effectively.

This study contributes to three strands of literature. The first is the literature on information frictions in agricultural value chains and rural markets (Anagol, 2017; Bai, 2021; Bergquist & Dinerstein, 2020; Bernard & Spielman, 2009; Bold et al., 2017; Mitra et al., 2018; Neupane et al., 2023; Roba et al., 2018). We contribute to this literature by focusing on information frictions between actors in the middle of the value chain, such as transporters and other intermediaries. This paper also contributes to the literature by providing insights into the behavioral responses of both farmers and transporters to the erosion of transporters' informational advantages. Analyzing the response of both farmers and transporters to the digital receipts allows us to understand how promoting transparency and observability affects the entire market. Second, this study contributes to a growing literature on how digital technologies can reduce information frictions and improve market functioning in agriculture (Aker & Mbiti, 2010; Fabregas et al., 2019; Jensen, 2007). While previous work has primarily emphasized digital tools that support farmer decisions at the production stage, our study focuses on improving observability and accountability in post-harvest transactions. By evaluating a simple intervention—such as SMS receipts—we show how digital tools can improve transparency in settings where monitoring is limited and formal enforcement mechanisms are weak. Lastly, due to the contract structure, this study also relates to the recent literature on relational contracts (Blouin & Macchiavello, 2019; Levin, 2003; Macchiavello, 2022; Morjaria & Macchiavello, 2015, 2020; Startz, 2019). The dairy supply chain in Uganda features informal, repeated interactions between farmers and transporters, making it a perfect setting for relational contracting. Information asymmetries within these relationships create space for opportunistic behavior. We contribute to this literature by investigating how the behavior of each actor, and the structure of the relational contract itself, shifts when one party gains access to verifiable information about the other's actions. More broadly, we contribute to this body of literature by identifying causal impacts of improved observability on product quality in an agricultural supply chain with informal contracts and documenting behavioral changes in relational contracts in response to reductions in information asymmetry.

In the next section, we describe the study context, including the structure of Uganda's dairy sector, seasonal production patterns, and the milk delivery process. Section 3 describes the digital receipts intervention. Section 4 presents the conceptual framework and outlines predictions about the effects of the digital receipt intervention across different types of farmers. Section 5 details the experimental design, randomization strategy, timeline of the intervention, data sources, and provides descriptive statistics. Section 6 outlines the empirical strategy used to estimate treatment effects across different margins. Section 7 presents the main results, including effects on delivery volumes, milk quality, and the mechanisms driving these outcomes. Section 8 concludes with broader implications, policy recommendations, and directions for future research.

2 Study Context: Uganda's Dairy Sector

The dairy supply chain in Uganda is characterized by many smallholder farmers that sell to local milk collection centers or dairy cooperatives (Van Campenhout et al., 2021, 2022). These cooperatives play a central role in the sector, particularly given the high degree of market fragmentation. By aggregating supply, cooperatives enable smallholder farmers to collectively access larger markets and negotiate better prices for their products. These prices fluctuate constantly depending on market conditions and overall supply, so cooperatives do not guarantee prices to farmers until the milk is sold. Cooperatives also pay members every two weeks in a lump sum without a breakdown of payments, limiting farmers' ability to clearly track the total amount delivered and the price paid on a given day. Beyond market access, cooperatives often provide a range of additional services, including veterinary care, livestock drugs, and financial services, making them a critical institution for rural livelihoods and an essential component of the Ugandan agricultural landscape.

The population of farmers targeted in this study is not homogeneous, they differ in livestock holdings and land ownership. Many face constraints such as poor road infrastructure, unreliable market information, and dependence on transporters. In this context, small-scale farmers tend to deliver for themselves while bigger farmers tend to rely on transporters to deliver their milk production to the cooperative. The digital receipt intervention aims to improve observability and transparency by sending SMS messages to farmers with daily delivery records. Our sample includes 766 farmers and 70 transporters across 13 dairy cooperatives in four districts in the Western Region of Uganda: Mbarara, Kiruhura, Isingiro, and Ibanda and the intervention was carried out over a three-month period.

2.0.1 Seasonality and Dairy Production in Uganda

Dairy production in Uganda is highly influenced by seasonal variation, particularly the alternation between rainy and dry periods. During the rainy season, pasture and water are more abundant, leading to better nutrition and higher milk yields. Cows have more access to quality forage, and the temperatures are less stressful for cows, both of which contribute to increased milk production. Conversely, during the dry season, pasture quality is lower, water sources become scarce, and cows often suffer from heat stress and nutritional deficits, leading to a decline in milk productivity.

Uganda experiences a bimodal rainfall pattern, with two rainy seasons and two dry seasons. In the western region of Uganda, the first rainy season typically spans from March to May, peaking in April, and the second occurs from September to November, with a peak in October and November. The driest months are June and July, with another relatively dry period in January and February. These patterns align with the agricultural calendar and directly affect forage availability and livestock productivity.

These seasonal dynamics are particularly consequential for small-scale farmers, many of whom operate with limited resources and small herds, primarily composed of a local breed: Ankole cows. While this breed is well-adapted to local climatic conditions, it is less productive in terms of milk output. A subset of these farmers—whom we refer to as *seasonal farmers*—either deliver only sporadic quantities of milk or stop delivering altogether during the dry season. In these months, the combination of pasture scarcity and limited water access leads to milk yields so low that market participation becomes unviable. Some farmers may instead use the little milk they produce for home consumption rather than sale.

As a result, pronounced seasonal fluctuations in milk deliveries to cooperatives are common in our setting, underscoring the importance of accounting for farmer heterogeneity when designing and evaluating interventions in the dairy value chain.

2.0.2 Milk Delivery Process

Milk delivery to the cooperative in Uganda follows a process that involves several steps. First, farmers (or transporters) bring fresh milk to the cooperative, where it is unloaded and prepared for inspection. Upon arrival, the milk goes through a quality and volume check—each batch is tested to ensure it meets the cooperative's quality standards. To discourage practices such as dilution with water, which artificially inflates volume while compromising quality, most cooperatives enforce a minimum specific gravity threshold to test for quality, typically measured using a lactometer[1]. In this setting, a reading of at least 26 degrees is often required to qualify for acceptance. Milk below this quality threshold is rejected by the cooperative. This quality control mechanism ensures that the milk collected and sold by the cooperative aligns with the standards demanded by downstream buyers, such as processors and traders.

Only milk that passes this quality threshold is accepted and formally recorded by the cooperative. Once accepted, the transaction enters a biweekly (two-week) payment cycle. During this period, milk deliveries are recorded daily, and payments are disbursed as a lump sum at the end of the cycle. Farmers are paid a uniform price per liter, regardless of milk quality, which can vary daily to reflect changes in market conditions or cooperative policies. Transporters typically receive a fixed rate of 100 UGX (about 0.03 USD) per liter—either for milk picked up when hired by a farmer or delivered to the cooperative when hired by the cooperative. Payment comes either from the cooperative, by deducting transportation costs

[1] As cooperatives grow, they often adopt milk analyzers, which provides additional quality metrics such as fat content.

from the farmer's earnings, for those contracted by the cooperative board, or directly from the farmer when the transporter is hired independently.

2.0.3 Mobile Cellphone Usage

As noted by Fabregas et al. (2019), the rapid spread of mobile phones in developing countries has created new opportunities to deliver timely and customized agricultural extension services at scale. In our study, the intervention was delivered via text messages (SMS), making access to a functional mobile phone a necessary condition for treatment. According to the 2023 Annual Communications Sector Report by the Ugandan Communication Commission, the penetration of mobile cellphones in the country is 81%. In other words, 8 out of every 10 Ugandans have a mobile cellphone. To ensure that all farmers in our sample were able to receive digital receipts, we conducted a phone validation survey in August 2024. Through this exercise, we successfully validated phone numbers for 688 out of the 766 farmers, representing 90% of our sample.

3 Digital Receipts Intervention

The intervention introduced an SMS-based digital receipts system to improve record keeping and transparency in milk transactions between farmers and dairy cooperatives. Each time a farmer delivered milk, either directly or through a transporter, the cooperative recorded the quantity and ran the quality tests. These records were then entered on a digital platform by our team. This information was then sent via SMS to the farmer's phone twice a week—every Monday and Thursday. The Monday message covered deliveries from the previous Thursday through Sunday, and the Thursday message covered deliveries from the previous Monday through Wednesday.

The content of the digital receipts included a personalized breakdown of delivery volumes and the name of the cooperative. For example:

> *"Your milk was delivered to [COOP NAME]. Here is a breakdown of your deliveries:*
>
> *2-SEPT: 24 L*
>
> *3-SEPT: 0 L*
>
> *4-SEPT: 19 L.*
>
> *Thank you!"*

The idea behind the digital receipts was simple: reduce information frictions in the supply chain. Without them, farmers get paid every two weeks in a lump sum, with no breakdown of daily deliveries or prices.

Combined with price fluctuations, this makes it hard for farmers to verify how much milk was recorded or detect discrepancies, especially when using a transporter. By giving farmers timely, itemized information, the digital receipts aimed to increase transparency, strengthen accountability, and change the way farmers interact with both transporters and cooperatives.

4 Conceptual Framework

Our study evaluates whether introducing digital receipts—SMS messages that report the quantity and timing of each delivery—can reduce information frictions between players at the early stage of the dairy value chain. In the Ugandan dairy sector, most farmers (regardless of how they deliver) must rely on payment amounts from cooperatives as a signal to estimate whether the milk they delivered was accurately recorded. Conceptually, digital receipts serve as a monitoring technology that enhances transparency between farmers, transporters, and cooperatives by improving the quality and timeliness of the information. This makes it easier for farmers to spot potential discrepancies, identify dishonest behavior, and take corrective action. By transforming previously unobservable actions into verifiable outcomes, digital receipts improve the signal associated with delivery behavior at the cooperative. This reduces moral hazard, strengthens accountability, enhances observability, and shifts the equilibrium of repeated farmer–transporter interactions toward more honest behavior.

We build a repeated game, modeling farmer-transporter relationships as long-term interactions where cheating may yield short-term gains for the transporter but risks future punishment (e.g., losing the client or being fired). Without monitoring, discrepancies are hard for farmers to detect. However, when digital receipts increase both the probability of detection and the potential benefit of switching away from a dishonest transporter, especially in a context of high competition among transporters and low switching costs, they make punishment strategies more credible. As a result, digital receipts improve the incentive compatibility of honest behavior.

On the other hand, consider farmers who do not use transporters and instead deliver milk to the cooperative themselves. For these farmers, the digital receipts intervention helps alleviate concerns about potential misconduct by cooperative staff, such as under-reporting deliveries. While such concerns are likely minimal for most self-delivery farmers, they may still arise since farmers do not have access to the cooperative's records, even when making their own deliveries. Moreover, these farmers make an active decision each day on both the extensive margin—whether to deliver at all—and the intensive margin—how much milk to deliver. In contrast, farmers who rely on transporters typically pre-arrange delivery schedules through informal agreements, specifying pickup times and days. As a result, their day-to-day decisions are limited (if anything) to the intensive margin, since the extensive margin decision (whether to deliver) has effectively been delegated in advance.

4.1 Hypothesis

Our framework suggests that among farmers who use transporters, we anticipate a reduction in under-deliveries, as digital receipts increase the likelihood that cheating, such as volume skimming, is detected, making honest behavior more incentive-compatible. In other words, we expect a greater volume of milk to be delivered on behalf of treated farmers, reflecting an improvement along the intensive margin. Additionally, treated farmers in this group should be more likely to detect discrepancies between what was picked up and what was delivered, and as a result, more likely to switch away from dishonest transporters, especially when switching costs are low and transporter competition is high. The effect on milk quality remains an empirical question, as it ultimately depends on how transporters respond to the increased transparency introduced by digital receipts and on who holds the upper hand in terms of bargaining power and enforcement capacity within the relational contract between farmers and transporters.

Finally, for farmers who deliver milk themselves, we anticipate that improved transparency may lead to changes on both the extensive and intensive margins. In other words, we would expect to see an increase in the likelihood of delivering on a given day, as well as greater quantities delivered when they do. That said, the transportation constraints faced by most self-delivering farmers may limit their ability to make large changes on the intensive margin. Therefore, we primarily expect to observe effects along the extensive margin.

We also expect treatment effects to materialize gradually, rather than immediately, due to a learning process. Farmers must first validate the accuracy of the digital receipts and confirm that the information they receive matches cooperative records. This verification process is likely to take between one and two payment cycles (roughly two to four weeks), depending on the frequency with which farmers deliver milk early in the intervention period.

5 Experimental Design

To evaluate the effects of reducing information frictions in the milk supply chain, we partnered with the Ugandan Dairy Development Authority (DDA) to implement a randomized controlled trial (RCT) across

13 dairy cooperatives in Western Uganda. The study covers 150 villages across four districts and includes 766 dairy farmers who regularly sell milk to their cooperative.

Farmers were randomly assigned to one of two groups:

- **Treatment Group** (378 farmers): Farmers received digital receipts (via SMS) detailing the volumes of milk recorded at the cooperative. Messages were delivered in RuNyankore –the local language– from a consistent caller ID[2].
- **Control Group** (388 farmers): Farmers continued under the status quo and did not receive any digital receipts.

Randomization was carried out at the farmer level and was stratified by:

1. Cooperative size
2. Whether the farmer was a *seasonal farmer*. As mentioned in Section 2, *seasonal farmers* are those who deliver only sporadic quantities of milk or stop delivering altogether during the dry season
3. Average daily milk volume delivered to the cooperative during the last payment period, conditional on delivering a positive amount

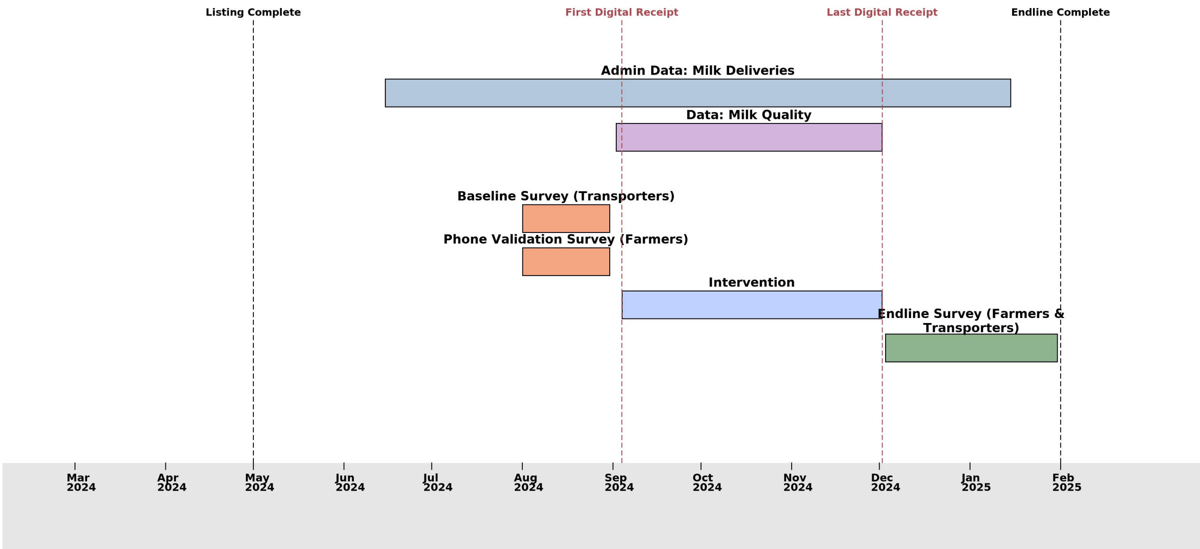
This stratification strategy aimed to ensure balance across key covariates and enhance statistical power to identify treatment effects. Given the level of randomization, potential spillovers could bias the treatment effect, they would bias our estimates downward. As a result, our findings should be interpreted as a lower bound on the true treatment effect.

5.1 Data Collection and Treatment Rollout

Baseline data collection took place in August 2024. These activities included a phone validation survey (completed for 683 farmers) and a baseline survey with 74 transporters. Treated farmers received an introductory message on September 4, 2024, informing them that they would begin receiving digital receipts twice per week. The first digital receipt was sent to treated farmers on Thursday September 5, 2024, and this receipt covered the deliveries done from Monday September 2 till Wednesday September 4, 2024 and continued for about three months. The last digital receipt was sent on December 2, 2024.

As mentioned in Section 3, farmers received digital receipts every Monday and Thursday: The Monday receipt covered milk deliveries recorded on the previous Thursday through Sunday while the Thursday messages covered deliveries made on the previous Monday through Wednesday.

[2]We acquired a caller ID through Uganda's two largest mobile service providers, Airtel and MTN, allowing all digital receipts to be sent from a unified sender ID labeled `DAIRY`, regardless of the service provider used by farmers.



In collaboration with the cooperatives in our sample and the Uganda Dairy Development Authority (DDA), we assembled a rich dataset combining administrative records, market prices, and survey data:

- **Administrative data** from cooperatives include daily volumes delivered, prices, payments, and the transporter associated with each delivery. These data span the pre-treatment, treatment, and post-treatment periods. To ensure accuracy, an enumerator visited each cooperative daily to verify that the digital records matched the cooperatives' records.

- **Survey data** include a phone validation survey and baseline in August 2024, and endline surveys in December 2024.

5.2 Data

We collected three main types of data. First, high-frequency administrative data was obtained from the cooperatives. These records include every delivery made by the farmers in our sample over a seven-month period, from June 15, 2024, to January 15, 2025. Additionally, using administrative data from the cooperatives, we identified the farmers that used transporters and the farmers that delivered their own milk. Second, during the treatment period, an enumerator visited one randomly selected cooperative from our sample each day and collected milk quality data using a lactometer from all farmers who delivered milk on that day. We use these enumerator-reported lactometer scores to examine whether the digital receipt intervention had any effect on milk quality. Lastly, we collected survey data from both farmers and transporters. Baseline surveys were conducted in August 2024. The farmer phone validation survey was conducted by phone with the primary goal of confirming the phone numbers of farmers in our sample. Once the team confirmed that a phone number belonged to a participating farmer, a short survey was administered. The survey covered questions on distance to the cooperative, transporter use, and milk production, in order to complement the administrative data used for randomization. The baseline survey of transporters was conducted in person at the cooperative where each transporter regularly delivers. The primary objective of this survey was to better understand each transporter's client base, daily delivery routes, socioeconomic background, and any additional income-generating activities they engage in. Lastly, we conducted in-person endline surveys with all transporters serving the cooperatives in our sample, as well as with a randomly selected subset of farmers across groups. These endline surveys captured changes in behavior and perceptions related to milk delivery, milk quality, transporter trustworthiness, and the use of digital receipts.

5.3 Descriptive Statistics

We begin by describing the key characteristics of both agents and principals in our study—transporters and farmers. The data combines information from our baseline transporter survey, administrative records from dairy cooperatives, and an endline survey of farmers.

5.3.1 Transporter Characteristics

Table 1 presents descriptive statistics from the baseline survey of 74 transporters operating across 13 cooperatives. About 32% of transporters in our sample are also dairy farmers. Transporters in our sample are, on average, 33 years old, with ages ranging from 18 to 74. Each transporter serves approximately 8.5 farmers, though this varies significantly, with some working with as few as one and others with up to 29 farmers. Transporters report making just over two trips per day[3], and 77% of them own their own motorcycle. Additionally, on average, each cooperative works with 4.7 transporters. These statistics highlight the heterogeneity in characteristics and the operational scale of milk transporters.

5.3.2 Farmer Sample and Baseline Balance

Our study includes 766 farmers randomly assigned at the individual level to treatment ($n = 378$) or control ($n = 388$), stratified by cooperative size, delivery volumes in the last payment period, and seasonal farmer status. Table 2 shows baseline balance checks using both administrative data and phone

[3] A trip is defined as a single drop-off at the cooperative, which may follow one or multiple farmer pickups. In other words, each delivery to the cooperative constitutes one trip, regardless of how many farmers were visited.

survey responses. There are no statistically significant differences between treatment and control groups in key baseline characteristics including validation of phone number, distance to the cooperative, average daily liters delivered, use of a transporter, or whether the transporter was hired by the cooperative. A joint F-test across these variables yields a p-value of 0.962, confirming that treatment and control groups are statistically comparable at baseline.

5.3.3 Farmer Demographics and Milk Production

Table 3 shows descriptive statistics from the endline survey of 396 farmers that were randomly selected from our sample and illustrates the socioeconomic and production characteristics of our sample. The average farmer is approximately 53 years old and lives in a household with about 7 members. They own an average of 10 productive cows and produce approximately 45 liters of milk per day. Of this, about 27 liters are sold to the cooperative, while the rest are either consumed at home or sold to other buyers such as private collection centers, traders, or local small businesses like restaurants.

On average, farmers sell to 1.2 buyers, indicating relatively limited marketing channels. About 62% of farmers report using a transporter to deliver milk to the cooperative. This figure is consistent with baseline administrative records and underscores the prevalence of transporter use in the dairy supply chain.

Together, these statistics offer a detailed snapshot of the key actors in our study and underscore the central role played by transporters in shaping farmer access to cooperative markets.

5.4 Milk Quality Data

During the treatment period, an enumerator visited one randomly selected cooperative from our study sample each day. At each visit, the enumerator recorded milk quality using a lactometer for every farmer who delivered milk to the cooperative that day. The lactometer reading is a gravity-based test that provides an index of water content in the milk. These measurements allow us to construct a consistent measure of milk quality over time. We use the enumerator-reported lactometer scores to estimate the effect of the digital receipt treatment on milk quality.

There are a couple of limitations to collecting milk quality data at the point of delivery rather than at the farm level. First, we are unable to observe milk quality when it left the farm, meaning our measure reflects quality after any tampering by the transporter may have occurred. Second, small-scale farmers often share transportation containers. To reduce costs, transporters typically collect and mix milk from multiple farmers at the farm level until the container is full. In such cases, only one lactometer reading was taken for the shared container, and this value was assigned to all farmers contributing to that delivery.

6 Empirical Strategy

6.1 Extensive Margin Analysis: Delivery Likelihood

To estimate the effect of the intervention on deliveries, we explore dynamic treatment effects on the extensive margin, the likelihood of making a delivery, by defining the outcome variable as a binary indicator equal to one if a farmer made a delivery on a given day, and zero otherwise. This design takes advantage of the seven months of panel data we collected on milk deliveries, allowing us to examine both pre-trends and dynamic treatment effects. We estimate an event study model using the following specification:

$$Y_{it} = \sum_{\substack{j=-7 \\ j\neq-1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \tau_j \cdot Treat_i + \alpha_i + \psi_t + \varepsilon_{it} \tag{1}$$

$$\begin{aligned} Y_{it} = & \sum_{\substack{j=-7 \\ j\neq-1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \tau_j \cdot Treat_i \\ & + \sum_{\substack{j=-7 \\ j\neq-1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \omega_j \cdot Treat_i \cdot SelfDeliver_i \\ & + \alpha_i + \psi_t + \sum_{\substack{j=-7 \\ j\neq-1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \lambda_j \cdot SelfDeliver_i + \varepsilon_{it} \end{aligned} \tag{2}$$

where Y_{it} denotes the outcome variable—whether or not they deliver—for individual i at time t, $\mathbb{1}\{t - t_i^* = j\}$ is an indicator for time relative to the last biweek prior to the intervention t_i^*. $Treatment_i$ is a binary indicator equal to one if individual i was part of the treatment group. The variable $SelfDeliver_i$ is a binary indicator equal to one for farmers who deliver their own milk at baseline (i.e do not use a transporter). The omitted group is $j = -1$, corresponding to the biweek immediately before the introduction of the digital receipts technology. The model includes individual fixed effects α_i and time fixed effects ψ_t, and an interaction between time and self-delivery status ($\sum_{\substack{j=-7 \\ j\neq-1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \lambda_j \cdot SelfDeliver_i$) to allow for differential time trends. Standard errors are clustered at the individual level, as treatment was randomized at that level.

6.2 Intensive Margin Analysis: Average Volumes Delivered to Cooperative

Similar to the extensive margin analysis, we implement an event study approach to explore how treatment effects evolve over time. Our identification strategy is as follows:

$$Y_{it} = \sum_{\substack{j=-7 \\ j\neq -1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \tau_j \cdot Treat_i + \alpha_i + \psi_t + \varepsilon_{it} \tag{3}$$

$$\begin{aligned} Y_{it} = &\sum_{\substack{j=-7 \\ j\neq -1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \tau_j \cdot Treat_i \\ &+ \sum_{\substack{j=-7 \\ j\neq -1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \omega_j \cdot Treat_i \cdot SelfDeliver_i \\ &+ \alpha_i + \psi_t + \sum_{\substack{j=-7 \\ j\neq -1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \lambda_j \cdot SelfDeliver_i + \varepsilon_{it} \end{aligned} \tag{4}$$

where Y_{it} denotes the outcome variable—average volumes delivered—for individual i at time t, $\mathbb{1}\{t-t_i^* = j\}$ is an indicator for time relative to the last biweek prior to the intervention t_i^*. $Treatment_i$ is a binary indicator equal to one if individual i was part of the treatment group. The variable $SelfDeliver_i$ is a binary indicator equal to one for farmers who deliver their own milk at baseline (i.e do not use a transporter). The omitted group is $j = -1$, corresponding to the biweek immediately before the introduction of the digital receipts technology. The model includes individual fixed effects α_i and time fixed effects λ_t, and an interaction between time and self-delivery status ($\sum_{\substack{j=-7 \\ j\neq -1}}^{7} \mathbb{1}(t - t_i^* = j) \cdot \lambda_j \cdot SelfDeliver_i$) to allow for differential time trends. Standard errors are clustered at the individual level, as treatment was randomized at that level.

6.3 Milk Quality Analysis

To explore the treatment effect on milk quality, we will use the following empirical strategy:

$$Y_{it} = \beta_1 \cdot Treat_i + \tau_s + \rho_t + \varepsilon_{it} \tag{5}$$

$$Y_{it} = \beta_1 \cdot Treat_i + \beta_2 \cdot Treat_i \cdot SelfDeliver_i + \tau_s + \rho_t + \varepsilon_{it} \tag{6}$$

In this specification, Y_{it} represents the outcome for individual i at time t, and $Treatment_i$ is a binary indicator for whether the individual was assigned to the treatment group. The model includes strata fixed effects τ_s to account for the stratified randomization design and month fixed effects ρ_t to control for temporal shocks common across cooperatives. Although treatment is randomized at the farmer level, milk quality was measured at cooperative visits where the enumerator recorded lactometer readings for all farmers delivering that day. To account for this design, the standard errors are clustered at the cooperative level using wild bootstrapping following Cameron et al. (2008).

6.4 Mechanisms Analysis

Lastly, to explore the potential mechanisms in place driving the results above, we use survey data and implemented the following empirical strategies:

$$Y_i = \rho_1 \cdot Treatment_i + \tau_s + \varepsilon_i \tag{7}$$

$$Y_i = \gamma_1 \cdot Treatment_i + \gamma_2 \cdot CoopHire_i + \gamma_3 \cdot (Treatment_i \cdot CoopHire_i) + \tau_s + \varepsilon_i \tag{8}$$

In these specification, Y_i represents the outcome for individual i. We focus on the following outcomes: (i) the farmer's ability to detect discrepancies between what they sent to the cooperative and what was actually delivered and recorded at the cooperative, (ii) whether the farmer switched from their baseline transporter after the intervention, and (iii) the level of trust the farmer has in their transporter after the intervention. $Treatment_i$ is a binary indicator for whether the individual was assigned to the treatment group. To better understand whether the way the transporter is hired plays a role in our outcomes, we add $CoopHire_i$ as a control, which is a binary indicator for whether the transporter was hired by the cooperative. The model includes strata fixed effects τ_s to account for the stratified randomization design. Standard errors are clustered at the individual level, as treatment was randomized at that level.

Together, these strategies allow us to estimate both average and dynamic treatment effects on milk deliveries, quality, delivery likelihood, and potential mechanisms. In the next section, we present the main empirical results from this analysis.

7 Results

This section presents the results of our statistical analysis. We begin by estimating the effect of the treatment on milk deliveries both at the extensive and intensive margin. Next, we assess the impact of the intervention on milk quality. Finally, we explore potential mechanisms driving these findings.

7.1 Extensive Margin Effects: Delivery Likelihood

n this section, we examine whether the treatment affected the frequency of milk deliveries, focusing on whether to deliver and thus capturing effects along the extensive margin. As described in Section 7, we used an event study approach. Figure 2 shows a statistically significant effect of the treatment on deliveries at the extensive margin for self-delivering farmers. In other words, treated farmers that self-delivered at baseline, on average, were more likely to deliver milk than control farmers that self-delivered at baseline during the study period. In terms of magnitude, treated self-delivering farmers were 5–9 percentage points more likely to deliver to the cooperative after the first biweek following the start of the treatment. Consistent with our hypothesis that treatment effects would materialize gradually through a learning process, we identify treatment effects within the first month of implementation. This finding is consistent with the hypothesis that digital receipts increase participation by improving transparency at the point of delivery. By enhancing trust in the accuracy of recorded transactions, the digital receipts may have mitigated perceived risks at the cooperative level, encouraging more consistent engagement among self-delivering farmers.

7.2 Intensive Margin Effects: Volumes Delivered to Cooperatives

In this section, we examine whether the treatment affected the volume of milk delivered to cooperatives, focusing on how much farmers delivered and thus capturing effects along the intensive margin. As with the analysis of extensive margin effects, we implement an event study approach using seven months of farmer-level delivery data. Figure 1 shows no statistically significant effect of the treatment on delivery volumes at the intensive margin, regardless of farmers' baseline delivery method. In other words, treated farmers did not, on average, deliver more milk than control farmers during the study period, whether they self-delivered or used a transporter at baseline.

One reason we may see no effect on volumes delivered is transportation constraints, regardless of baseline delivery method. Farmers who self-deliver typically transport milk by walking or bicycle, which physically limits how much they can carry. Farmers who use transporters face a similar, though different, constraint:

the volumes they send to the cooperative must be arranged in advance, and increases are only possible if the transporter has the capacity to carry the additional liters. In Figure 14 and Figure 15, we observe *bunching* at different levels, regardless of the farmer's delivery method. These patterns highlight potential transportation constraints that may limit farmers' ability to increase volumes delivered to cooperatives.

7.3 Milk Quality

In this section, we examine whether the treatment impacted the quality of milk delivered to cooperatives, as measured by lactometer readings. Higher readings indicate greater density and thus lower water content, serving as a proxy for reduced dilution.

For this outcome, as shown in Table 4, we estimate an OLS specification leveraging the randomized design. In Table 5, we also examine heterogeneity by transporter usage to assess whether the treatment had differential effects based on farmers' reliance on transporters. This heterogeneity analysis helps identify where the behavioral change in milk quality is coming from—whether from farmers or from transporters.

We find that the treatment led to a statistically significant improvement in milk quality. On average, treated farmers delivered milk with lactometer readings that were 0.1 degrees higher than those of control farmers. While this may appear modest in magnitude, it represents a significant improvement given the market conditions of the Ugandan dairy sector. To put this in context, the average positive effect of the rainy season on milk quality in our data is 0.78. Framed relative to seasonal variation, the intervention's impact amounts to nearly 13% of the average quality gain observed during the rainy season, underscoring its substantive effect on milk quality.

To test our theoretical prediction and better understand where the positive effects on quality are coming from, we match farmers and transporters at baseline and compare them to the endline matches. We then identify all farmers using transporters at baseline and compare the milk quality of those who switched their baseline transporter (*switchers*) versus those who kept the same transporter (*keepers*). Tables 6 and 7 present the results of this analysis. While we are underpowered to make any causal claims, the coefficients for *switchers*, treatment, and the interaction term for treated farmers who also switched transporters are all positive in the first column. This pattern suggests that the gains in milk quality among farmers using transporters come primarily from those who switched away from their baseline transporter, especially treated farmers who made the switch.

By increasing the cost of dishonest behavior, the treatment seems to have encouraged higher-quality deliveries. These gains are concentrated among farmers who use transporters and switched their baseline transporter, reinforcing the idea that the intervention helped farmers identify misbehaving transporters and limited opportunities for tampering with milk during transport by making transporters' actions more observable.

7.4 Mechanisms

The results presented above are consistent with a set of behavioral mechanisms rooted in improved transparency and accountability introduced by digital receipts. These mechanisms are particularly relevant in contexts where informational asymmetries between farmers and transporters previously allowed for opportunistic behavior. We identify three main channels supported by complementary evidence from survey data.

7.4.1 Discrepancy Detection

Table 8 shows that treated farmers using transporters were significantly more likely to detect discrepancies between the volumes picked-up and those recorded at the cooperative. Among farmers who relied on transporters, those in the treatment group were 20–26 percentage points more likely to report inconsistencies in delivery records. This suggests that digital receipts enhanced observability, allowing farmers to cross-check cooperative records against their own expectations or the volumes they handed off. This improved ability to monitor deliveries likely increased the perceived risk of misreporting and siphoning by transporters.

7.4.2 Transporter Switching

Table 9 shows that treated farmers using transporters were 14–17 percentage points more likely to switch transporters over the course of the intervention. Access to verifiable information reduced the cost of assessing transporter performance and increased the perceived benefit of moving away from under-performing or dishonest transporters. This points to a disciplining mechanism: s farmers gained timely and more accurate information, they were better positioned to respond strategically to misconduct, using the threat of exit as a tool to enforce accountability and deter misconduct

7.4.3 Trust Decline

Table 10 shows that treated farmers using transporters expressed lower levels of trust in their current transporter relative to untreated farmers. This reduction in trust may reflect increased awareness of misbehavior that was previously concealed by information frictions. Notably, the largest decline in trust occurred among farmers who hired their own transporter (often a friend or relative) suggesting that digital receipts revealed misbehavior even in relationships with high baseline trust. This reinforces the interpretation that the intervention made misconduct more visible, prompting farmers to reassess the reliability of their transporters and adjust their behavior accordingly.

7.5 Individual Behavior and Market Dynamics

Since we identify effects at the extensive margin in deliveries for self-delivery farmers, but not at the intensive margin—our results suggest that the treatment worked more as a behavioral nudge than as a tool for resolving cooperative-level information frictions, which are minimal for these farmers.

We also find a modest increase in the share of farmers using transporters at endline compared to baseline. At the start of the intervention, approximately 63% of farmers in our sample relied on transporters to deliver their milk to cooperatives. By the end of the intervention, that share had risen to nearly 66%. This change reflects underlying dynamics: 46 farmers stopped using transporters, while 68 began using them. At the same time, the number of transporters offering services to the cooperatives in our sample increased from 74 to 81. While this is not a large increase, the fact that such a shift occurred over a relatively short period suggests that farmers may adjust their delivery method by leveraging the monitoring provided by digital receipts to reassess the costs and benefits of using transporters.

Together, the evidence suggests that digital receipts shifted behavior primarily by increasing the visibility of transactions and empowering farmers to act on that information—especially in relationships characterized by weak accountability or low transparency. Rather than uniformly improving outcomes across all market actors, the effects of the intervention were more pronounced where pre-existing information frictions were severe: in transactions involving transporters. These results highlight how the effectiveness of digital interventions depends critically on the structure of market relationships and existing information frictions. Even simple, low-cost digital tools that improve information flow can lead to meaningful changes in behavior by making it easier to monitor transactions or simply helping farmers see what was previously hidden.

7.6 Potential Spillovers

We anticipate two key channels through which spillover effects may occur, both of which could bias our treatment estimates downward. The first channel is on the demand side, through farmer-to-farmer information sharing. Treated farmers who receive digital receipts may become aware of discrepancies between what they sent and what was delivered (and reported) at the cooperative, revealing dishonest behavior by their transporter. Because farmers often rely on shared transporters and maintain strong social ties within their communities, treated individuals may share their experiences with peers, especially when trust or misconduct is at stake. As a result, untreated farmers who use the same transporter could be indirectly affected by the treatment, either by gaining information about the transporter's behavior or by adjusting their own delivery practices in response to concerns about potential misconduct.

The second channel is on the supply side, from changes in transporter behavior after the treatment started. Transporters may become aware that some farmers are receiving digital receipts but may not know exactly who is treated. In response, they could choose to adjust their behavior toward all farmers in order to minimize the risk of detection. This uncertainty could induce more honest behavior across the board, even among untreated farmer–transporter pairs.

These mechanisms allow control farmers to benefit indirectly from the treatment, either through increased information or improved intermediary behavior, the estimated treatment effects will likely understate the true impact of the treatment. In addition, because only one lactometer reading was collected for each shared container and assigned to all contributing farmers, the estimated treatment effect on milk quality is likely biased downward. Overall, these factors suggest that our estimates should be interpreted as a lower bound of the true treatment effect on farmer outcomes.

8 Discussion and Conclusion

This paper provides causal evidence on how simple digital technologies can reduce information frictions and improve outcomes in agricultural markets. We study a randomized intervention in western Uganda that sent digital receipts via SMS to farmers, reporting the quantity of milk delivered on their behalf to the cooperative each day. Our results show that digital receipts led to a statistically significant improvement in milk quality, as measured by lactometer readings. While not statistically significant, we also find that farmers who switched from their baseline transporter tend to deliver less diluted milk, suggesting that the treatment improved the overall pool of reliable transporters in the market. In other words, greater observability allowed farmers to detect and avoid misbehaving transporters, shifting the composition of transporters and contributing to an overall reduction in milk dilution. While farmers do not receive higher prices for better quality milk, the reduction in dilution may carry important public health benefits, especially in contexts where the water used to adulterate milk is unsafe, as is often the case in rural Uganda.

Indeed, improvements in milk quality may have important public health implications. Anderson et al. (2022) document that milk inspections in late 19th-century U.S. cities reduced mortality from waterborne diseases by 12–19%. In rural Uganda, where access to potable water is limited, water used for dilution is likely drawn from natural sources or untreated tap water. This raises concerns about introducing parasites and pathogens into the milk supply, particularly for vulnerable populations such as children and immunocompromised individuals (Sente et al., 2023). These risks are especially important given that cooperatives in our study area are key suppliers of milk to households and small businesses in nearby rural villages. More broadly, quality upgrading is key for economic growth, especially low- and middle-income countries (Verhoogen, 2023), making our findings relevant not only for farmer welfare and public health, but also for longer-term development.

We also find that digital receipts increased delivery consistency among self-delivering farmers. Treated self-deliverers were more likely to deliver during the treatment period and continued to do so even after the intervention ended, including during the dry season when delivery volumes typically decline. This has important implications for market access and sustainability, especially for small dairy cooperatives in Uganda. These cooperatives often depend on small-scale farmers with limited productive capacity and operate with thin margins. During the dry season, when milk yields drop, cooperatives frequently struggle to meet buyer demand. In this sense, digital receipts functioned not only as a monitoring tool but also as a behavioral nudge, encouraging more regular participation by smallholder farmers and supporting the viability of small dairy cooperatives.

Together, these findings show that even low-cost technologies can improve observability and accountability, reduce strategic behavior, and support more consistent engagement in cooperative markets. As digital infrastructure expands in rural areas, tools like SMS receipts may offer a scalable way to strengthen smallholder supply chains, especially in settings with limited enforcement and high monitoring costs.

At the same time, our results highlight the importance of physical and logistical constraints that may limit how farmers respond to improved monitoring. Self-delivering farmers are constrained by how much they can carry on foot or by bicycle, while those relying on transporters depend on the transporter's carrying capacity and pre-arranged schedules. These limitations may partially explain why we do not observe changes at the intensive margin in delivered volumes, even as observability and accountability improved.

Future research could examine how digital receipts interact not only with price incentives or alternative contracts, like premiums for higher-quality deliveries, but also with efforts to alleviate transportation constraints. Pairing these approaches could strengthen observability and accountability while aligning incentives among farmers, transporters, and cooperatives. Another open question is the long-term adoption and sustainability of digital receipt systems, particularly when implemented by cooperatives or processors themselves.

Overall, these insights highlight the potential for simple, low-cost digital technologies to improve transparency, strengthen accountability, and shift behavior in fragmented agricultural markets. By reducing information asymmetries and reinforcing daily participation decisions, digital receipts can deliver meaningful improvements in supply chain integrity and farmer engagement, even without direct financial incentives. As digital infrastructure continues to expand across Sub-Saharan Africa, interventions like these offer a scalable and cost-effective way to improve the functioning of informal markets and strengthen smallholder-oriented institutions.

9 Figures

Figure 1: Event Study on Average Volumes Delivered

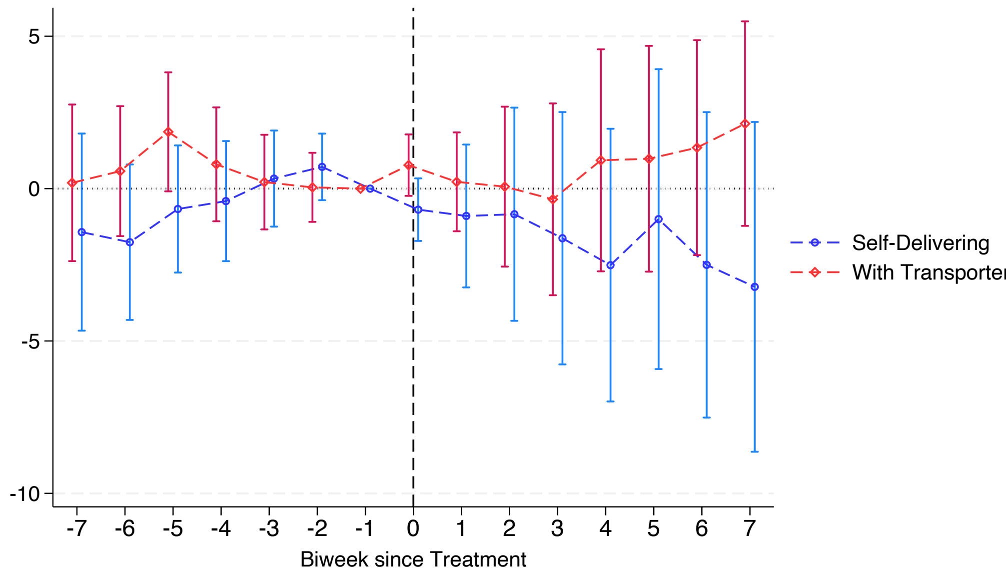
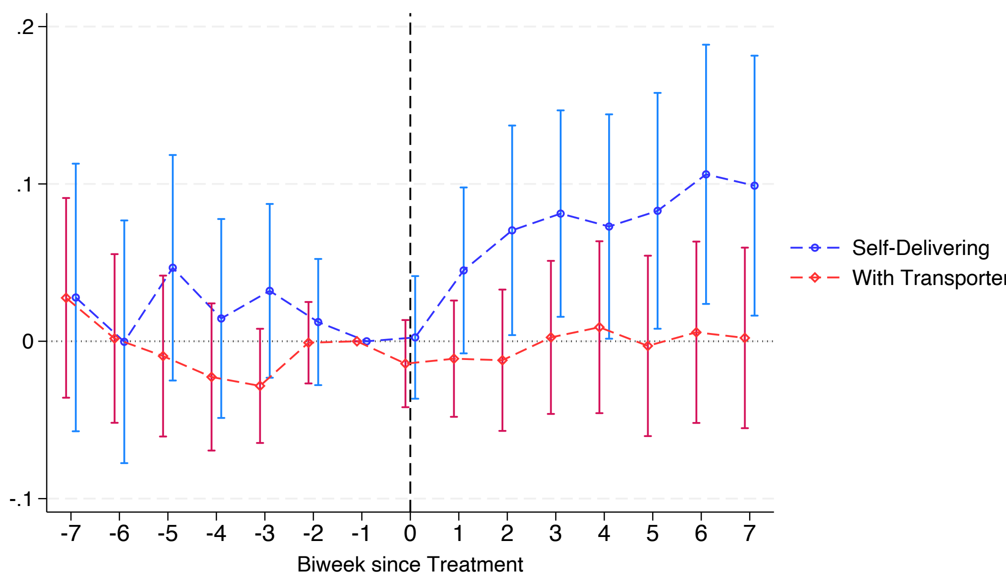


Figure 2: Event Study on Delivery Likelihood



10 Tables

Table 1: Transporter Baseline Characteristics

Variable	Mean	Min	Max	*N*
Age	33.44	18	74	74
Farmers per transporter	8.46	1	29	74
Trips per day	2.19	1	8	74
Own motorcycle (1=yes)	0.77	0	1	74
Transporters per cooperative	4.71	1	13	74

Table 2: Baseline Characteristics and Balance

	Control		Treatment vs Control		Farmers
	Mean	Std. dev.	Coeff.	Std. err.	*N*
	(1)	(2)	(3)	(4)	(5)
Validate phone number	0.894	(0.308)	0.008	(0.022)	766
Distance to coop (minutes)	18.320	(10.493)	0.399	(0.746)	766
Number of liters delivered	16.577	(28.562)	0.050	(2.005)	766
Uses transporter	0.619	(0.486)	0.019	(0.035)	766
Coop hires transporter	0.571	(0.496)	-0.009	(0.045)	484
P-value (joint F-test)				0.962	

Table 3: Farmer Endline Characteristics

Variable	Mean	SD	Min	P25	P75	Max
Age	52.90	15.02	19	42	63	94
Household members	7.29	3.52	1	5	9	20
Productive cows	10.42	9.01	0	4	15	60
Milk production (L/day)	44.71	45.81	2	16	55	400
Liters sold to coop (L/day)	26.57	29.69	0	8	35	202
Number of buyers	1.24	0.63	0	1	1	9
Uses transporter service (1=yes)	0.62	—	—	—	—	—

Table 4: Effect of Treatment on Average Quality

	Overall	Month 1	Month 2	Month 3
Treatment	0.100*	0.108	0.091	0.121**
	(0.053)	(0.105)	(0.093)	(0.052)
Strata FE	Yes	Yes	Yes	Yes
Month FE	Yes	No	No	No
Observations	1419	472	498	449
Control mean	27.535	27.180	27.492	27.957
R^2	0.459	0.284	0.518	0.708

Standard errors are clustered at the cooperative level.
Computed using wild cluster bootstrapping.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Effect of Treatment on Average Quality, by Transporter Usage

	Overall	Month 1	Month 2	Month 3
Treatment	0.183	0.359**	0.105	0.045
	(0.112)	(0.138)	(0.148)	(0.150)
Transporter	-0.150	-0.186	-0.144	-0.112
	(0.094)	(0.227)	(0.091)	(0.078)
Treatment × Transporter	-0.128	-0.387	-0.024	0.100
	(0.159)	(0.241)	(0.223)	(0.152)
Strata FE	Yes	Yes	Yes	Yes
Month FE	Yes	No	No	No
Observations	1419	472	498	449
Control mean	27.535	27.180	27.492	27.957
R^2	0.462	0.297	0.519	0.707

Standard errors are clustered at the cooperative level.
Computed using wild cluster bootstrapping.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Effect of Treatment on Average Quality, Switchers versus Keepers I

	Overall	Month 1	Month 2	Month 3
Switcher	0.063	0.025	0.115	0.032
	(0.089)	(0.193)	(0.149)	(0.058)
Strata FE	Yes	Yes	Yes	Yes
Month FE	Yes	No	No	No
Observations	907	290	306	311
Control mean	27.578	27.100	27.523	28.078
R^2	0.538	0.381	0.546	0.742

Standard errors are clustered at the cooperative level.
Computed using wild cluster bootstrapping.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Effect of Treatment on Average Quality, Switchers versus Keepers II

	Overall	Month 1	Month 2	Month 3
Switcher	0.006	0.055	-0.054	0.053
	(0.105)	(0.214)	(0.178)	(0.088)
Treatment	0.022	0.036	-0.047	0.163***
	(0.089)	(0.185)	(0.107)	(0.038)
Switcher × Treatment	0.122	-0.059	0.347*	-0.035
	(0.101)	(0.214)	(0.180)	(0.120)
Strata FE	Yes	Yes	Yes	Yes
Month FE	Yes	No	No	No
Observations	907	290	306	311
Control mean	27.578	27.100	27.523	28.078
R^2	0.538	0.377	0.548	0.744

Standard errors are clustered at the cooperative level.

Computed using wild cluster bootstrapping.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Effect of Treatment on Spot Discrepancy (LPM)

	(1) LPM: Spot Disc.	(2) LPM: Spot Disc.
Treatment	0.196***	0.257***
	(0.055)	(0.097)
Coop hires transporter		0.184
		(0.120)
Treatment × Coop hires		-0.096
		(0.118)
Strata FE	Yes	Yes
Observations	245	245
Control mean	0.181	0.181
R^2	0.215	0.223

Only farmers using transporters. Standard errors are clustered at the farmer level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Effect of Treatment on Transporter Switching (LPM)

	(1) LPM: Change Transp.	(2) LPM: Change Transp.
Treatment	0.141***	0.174***
	(0.037)	(0.054)
Coop hires transporter		0.038
		(0.059)
Treatment × Coop hires		-0.054
		(0.075)
Strata FE	Yes	Yes
Observations	246	246
Control mean	0.034	0.034
R^2	0.248	0.250

Only farmers using transporters. Standard errors are clustered at the farmer level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Effect of Treatment on Trust in Transporters (LPM)

	(1) LPM: Trust	(2) LPM: Trust
Treatment	-0.118*	-0.221**
	(0.063)	(0.095)
Coop hires transporter		-0.427***
		(0.123)
Treatment × Coop hires		0.156
		(0.126)
Strata FE	Yes	Yes
Observations	246	246
Control mean	0.698	0.698
R^2	0.154	0.194

Only farmers using transporters. Standard errors are clustered at the farmer level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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11 Appendix

Figure 3: Uganda—Geographical Location of the Study I

Figure 4: Uganda—Geographical Location of the Study II

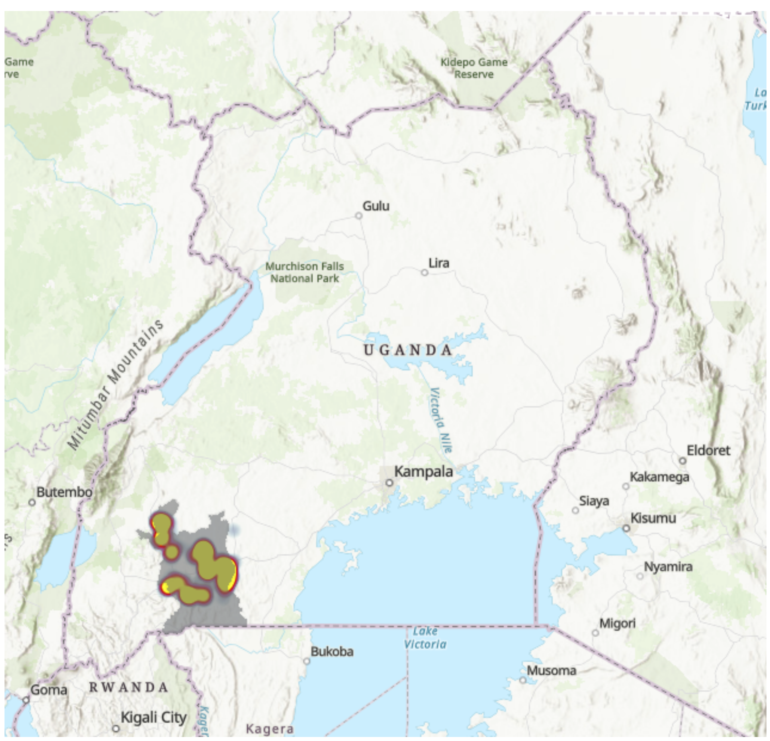

Figure 5: Western Region—Geographical Location of the Study I

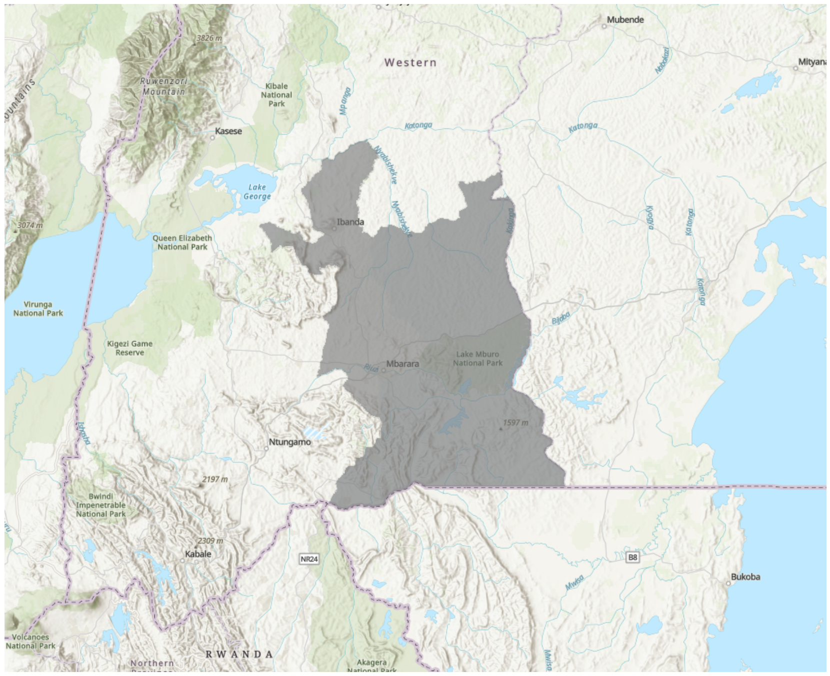

Figure 6: Western Region—Geographical Location of the Study II

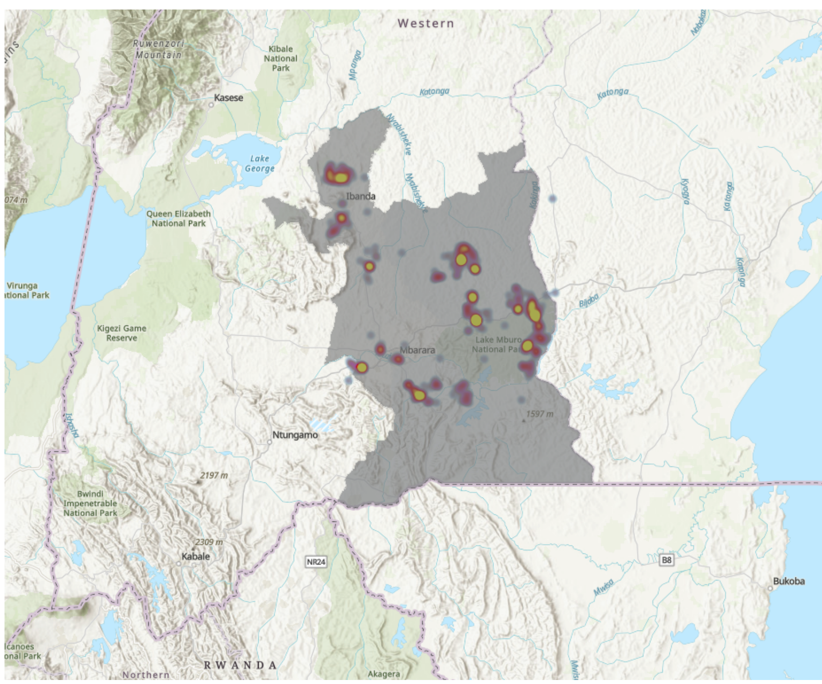

Figure 7: Dairy Cooperative

Figure 8: Transporter

Figure 9: Milk Testing—Quality Control

Figure 10: Digital Receipts—Example

Figure 11: Average Volumes Delivered per Group (Monthly)

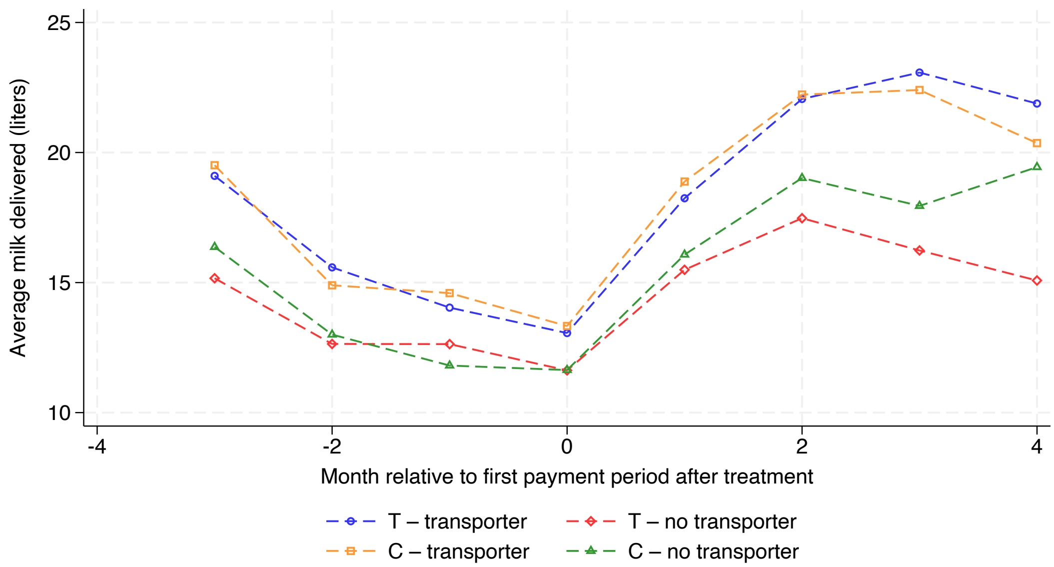

Figure 12: Average Log Volumes Delivered per Group (Monthly)

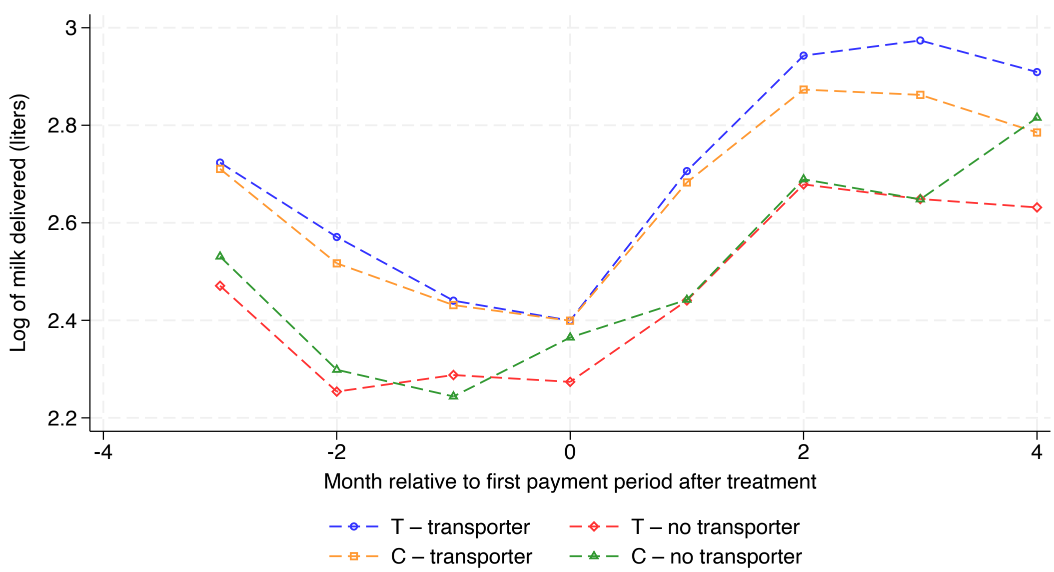

Figure 13: Probability of Delivery per Group (Monthly)

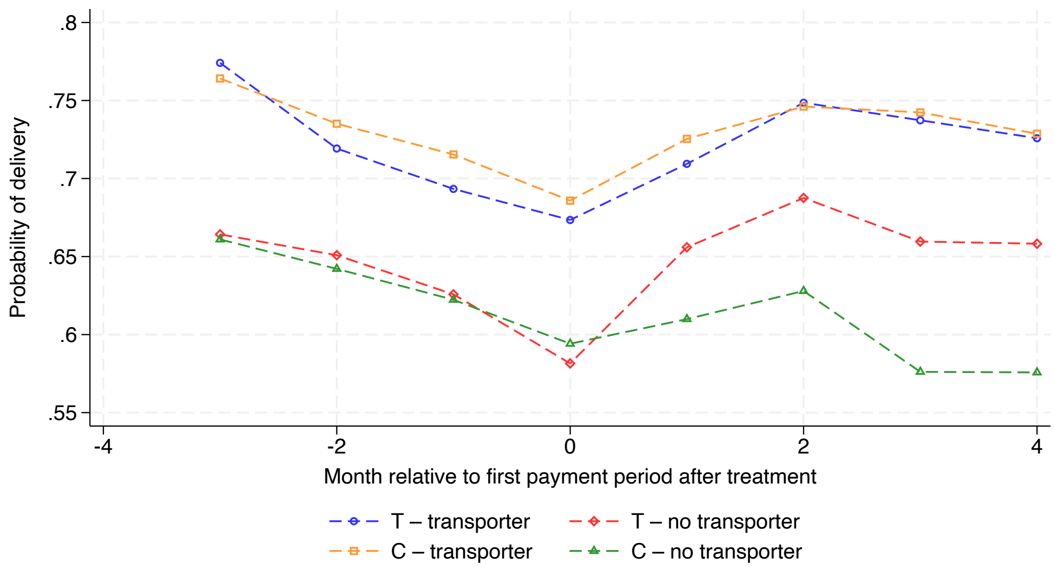

Figure 14: Histogram on Administrative Deliveries – Self-Delivering Farmers

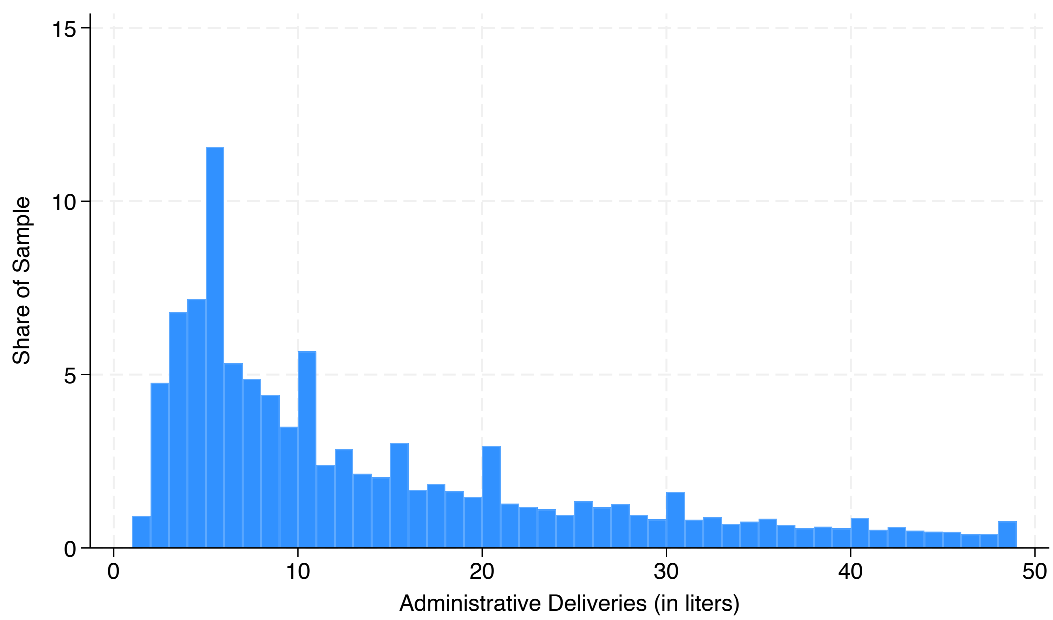

Figure 15: Histogram on Administrative Deliveries – Farmers using Transporters